

Lab 1: Basic Logic Design

Requirements Not Met

N/A

Problems Encountered

No Major Problems in the lab, misread the truth table a couple of times but fixed my mistakes.

Applications

There are many applications for the Boolean logic and basic circuit designs seen in this lab. One example is the very circuit designed in part 3, which is a basic alarm system that takes in multiple inputs to give out multiple outputs depending on the digital logic. Another example could be using daylight sensors and light switches to automatically turn outdoor lights on and off during the night. Many other household functions and simple machinery depends on the simple logic design taken in this lab.

Pre-Lab Questions

Lab 1-part 1 Boolean Algebra Simplification

a) $F = A \cdot B + A \cdot \bar{B} + \bar{A} \cdot B$ $A \cdot B + A \cdot \bar{B} = A(B + \bar{B})$
 $F = A(B + \bar{B}) + \bar{A} \cdot B$ $B + \bar{B} = 1$
 $F = A + \bar{A} \cdot B$ $A + BC = (A + B)(A + C)$
 $(A + \bar{A})(A + B)$
 $(A + 1)(A + B)$
 $F = A + B$

b) $Z = \bar{V}W + X + \bar{Y}(X + \bar{V}W)$
 $(1 + \bar{Y})(X + \bar{V}W)$
 $Z = X + \bar{V}W$

c) $A(D, W, M)$

D	W	M	A	S
0	0	0	0	1
0	0	1	1	1
0	1	0	0	0
0	1	1	1	1
1	0	0	0	1
1	0	1	0	0
1	1	0	0	0
1	1	1	0	1

$A = M\bar{D}\bar{W} + MW\bar{D} + MD\bar{W}$
 $X = W(\bar{D}\bar{W} + W\bar{D} + D\bar{W})$
 $Y = M(W\bar{D} + \bar{W}(D + \bar{D}))$
 $M(W\bar{D} + \bar{W}1)$
 $M(W\bar{D} + \bar{W}) = M(\bar{W} + \bar{W}) = M(\bar{W} + \bar{W})$
 $M(\bar{W} + \bar{W})(\bar{W} + \bar{D})$
 $A = M(\bar{W} + \bar{D})$





